

Contralateral-Normalized Deformation after Chronic Stroke: Cross-sectional Description and Aphasia Prediction in ARC

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Abstract

Purpose: Nonlinear registration fields around chronic stroke lesions contain anatomical signal but also premorbid asymmetry, tissue collapse, and registration error. We developed a contralateral-normalized lesion-associated deformation proxy and tested whether it improves held-out prediction of Western Aphasia Battery Aphasia Quotient (WAB-AQ) beyond conventional lesion features. **Materials and Methods:** From the Aphasia Recovery Cohort, we analyzed one T1-weighted acquisition per participant. An affine component fitted in the contralesional hemisphere was removed from the SVReg inverse coordinate map. The residual was contrasted with its mirrored contralesional homolog and retained only in the lesional hemisphere. Lesion and inpainting-target voxels were excluded. Magnitude, radial displacement, and log-Jacobian asymmetry were summarized 3–20 mm from the lesion. Ridge models were evaluated by 5-fold nested cross-validation repeated 20 times on identical splits. **Results:** Of 214 deformation cases, 211 matched the clinical table and one failed laterality quality control, leaving 210 participants. Median near-lesion magnitude was 6.20 mm (IQR 5.32–7.84). The conventional lesion model achieved mean absolute error (MAE) 16.42 AQ points; adding deformation yielded 16.04. The participant-level mean MAE advantage was 0.38 points (95% bootstrap CI -0.50 to 1.17). Adding deformation after lesion-uncertainty features gave -0.06 points (CI -0.77 to 0.61). **Conclusion:** The proxy provides an auditable cross-sectional description of lesion-adjacent registration geometry, but its incremental predictive benefit was not statistically robust because the primary mean-effect interval included zero. It must not be interpreted as pressure or physical ground-truth mass effect.

1 Introduction

Spatial normalization of brains with focal lesions is difficult because an intensity-driven transformation may deform normal anatomy to reduce mismatch at the lesion [1, 2]. Lesion masking and filling can reduce this bias [3], but the resulting transformation remains a mixture of global pose, natural anatomical variation, lesion-associated remodeling, and algorithmic error. These qualifications are especially important in chronic stroke, where lesions and whole-brain morphology can continue to evolve years after the event [4].

Deformation-based morphometry uses displacement and transformation Jacobians to describe local shape or volume differences [5]. Tumor studies have related modeled deformation to radiological mass-effect scores [6], but that mechanistic interpretation does not transfer directly to chronic infarction. Without a premorbid or acute reference scan, a single chronic image cannot identify the causal displacement produced by the stroke. A useful cross-sectional descriptor must therefore separate its measurable geometric definition from an unidentifiable physical claim.

The Aphasia Recovery Cohort (ARC) provides a large, openly released chronic stroke imaging and behavioral resource [7]. Lesion anatomy is a strong predictor of aphasia severity, while extralesional

morphometry may contain additional information [8]. We asked whether a compact set of lesion-adjacent deformation summaries adds held-out predictive value beyond age, assessment delay, lesion size, laterality, and language-network lesion burden. Our contributions are: (1) a within-participant, contralateral-normalized field with exact lesional-only support; (2) explicit registration-sensitivity and folding quality controls; and (3) a leakage-resistant, repeated nested-CV evaluation designed to support a valid negative result.

2 Materials and Methods

2.1 Data and cohort construction

ARC release 1.0.2 contains de-identified BIDS imaging and participant-level behavioral variables from chronic stroke survivors [7]. WAB-AQ summarizes spontaneous speech, comprehension, repetition, and naming on a 0–100 scale, with lower values indicating greater impairment [9]. The source dataset reports informed consent under University of South Carolina institutional review and exemption of the anonymized public release from further review [7].

The analysis used one acquisition per participant from the case table paired to the participant’s WAB record. The deformation manifest contained 214 unique cases. Completed deformation, clinical, and optional uncertainty tables were joined one-to-one by acquisition identifier. Multiple acquisitions per participant after joining were prohibited. The final cohort had median age at stroke 58.5 years (IQR 50.0–66.0), WAB assessment delay 721 days (IQR 404–1658), WAB-AQ 69.1 (IQR 40.8–90.0), and lesion volume 79.1 mL (IQR 34.5–154.7). Lesions were left-dominant in 209 cases and right-dominant in 1.

Upstream lesion masks were produced by a full-resolution nnU-Net [10] trained on ATLAS v2 [11]. Inpainted T1 images were used only to provide a continuous substrate for BrainSuite v23a processing [12] and surface-constrained volumetric registration (SVReg) [13]. No generated intensity, label, or morphometric measurement inside the lesion or dilated inpainting target was treated as biological data.

2.2 Contralateral-normalized deformation

Let $F(\mathbf{x})$ be the SVReg inverse coordinate map giving the subject-voxel coordinate corresponding to atlas voxel $\mathbf{x}$. On valid contralesional brain voxels more than two voxels from the midline, we fitted

$$F_A(\mathbf{x}) = M\mathbf{x} + \mathbf{b} \quad (1)$$

by least squares with five iterations of median-absolute-deviation outlier rejection. The affine-removed residual, expressed in atlas-axis millimeters, was

$$\mathbf{u}(\mathbf{x}) = M^{-1}\{F(\mathbf{x}) - F_A(\mathbf{x})\}. \quad (2)$$

Masked normalized Gaussian smoothing used a 2-mm kernel.

Let R reflect position through the symmetric atlas midline and reverse a vector’s left–right component. The lesion-associated asymmetry field was

$$\mathbf{u}_L(\mathbf{x}) = \mathbf{u}(\mathbf{x}) - R\mathbf{u}(R\mathbf{x}). \quad (3)$$

Both members of every mirrored pair had to lie within valid atlas and mapped subject brain masks. The lesion, the 3-mm-dilated inpainting target, and its mirror were excluded. Equation 3 was

retained only in the lesional hemisphere; all values in the reference hemisphere were stored as exact zeros.

Magnitude was $m = \|\mathbf{u}_L\|$. The gradient of the Euclidean distance outside the lesion defined outward unit normal $\mathbf{n}$, and radial displacement was $r = \mathbf{u}_L \cdot \mathbf{n}$ (positive outward, negative inward). Local volume change was $J = \det(I + \nabla \mathbf{u})$. For positive paired Jacobians we calculated $\Delta \log J = \log J(\mathbf{x}) - \log J(R\mathbf{x})$. Summaries were retained in 3–5, 5–10, 10–20, and 20–40 mm shells.

2.3 Quality control and sensitivity

Eligible cases required lesion laterality index $|V_L - V_R|/(V_L + V_R) \geq 0.80$, nonpositive-Jacobian fraction no greater than 0.05, and at least 1,000 valid voxels 3–20 mm from the lesion. The direct, non-inpainted SVReg field was normalized identically when available. The magnitude of the inpainted-minus-direct field difference was recorded as registration sensitivity, not as biological deformation.

The predictive analysis used eight prespecified 3–20 mm summaries: median and 95th-percentile magnitude; median and mean-absolute radial displacement; outward and inward radial integrals; and positive and negative log-Jacobian integrals. A sensitivity model additionally included median registration sensitivity and contralateral affine-fit RMSE.

2.4 Aphasia prediction and inference

The clinical model contained age at stroke and $\log(1 + \text{WAB days})$. The conventional lesion model added total lesion volume, left and right language-network lesion burden, and lesion laterality. The primary incremental model added the eight deformation summaries. Optional models added soft-lesion and boundary-ambiguity features, with and without deformation.

Within each training fold, missing predictors were median-imputed, standardized, and fit by ridge regression. A four-fold inner loop selected the penalty from 17 log-spaced values between 10^{-4} and 10^4 by MAE. A five-fold shuffled outer loop was repeated 20 times using identical splits for all models. Repeating every preprocessing and selection step inside the outer loop limits model-selection bias [14].

Performance measures were out-of-fold MAE, RMSE, R^2 , and Pearson r . The primary effect was each participant’s absolute error under the conventional model minus error under the deformation model, averaged over repeats; positive values favor deformation. A 5,000-resample participant bootstrap estimated its 95% CI. Two-sided paired Wilcoxon tests were Holm-adjusted across comparisons with the conventional model [15]. Because the signed-rank test and the bootstrap mean interval address different estimands, predictive superiority required the prespecified mean-advantage interval to exclude zero.

3 Results

3.1 Cohort flow and deformation characteristics

Of 214 unique deformation cases, 211 matched the clinical table. One bilateral case failed laterality QC, leaving 210 participants for every model. Median nonpositive-Jacobian fraction was 1.49% (IQR 1.30–1.66%).

Across participants, median 3–20 mm magnitude was 6.20 mm (IQR 5.32–7.84), median radial displacement was 0.16 mm (IQR -0.67 – 0.99), and mean-absolute radial displacement was 3.53 mm

M2238: lesion-associated deformation on the acquisition-native T1

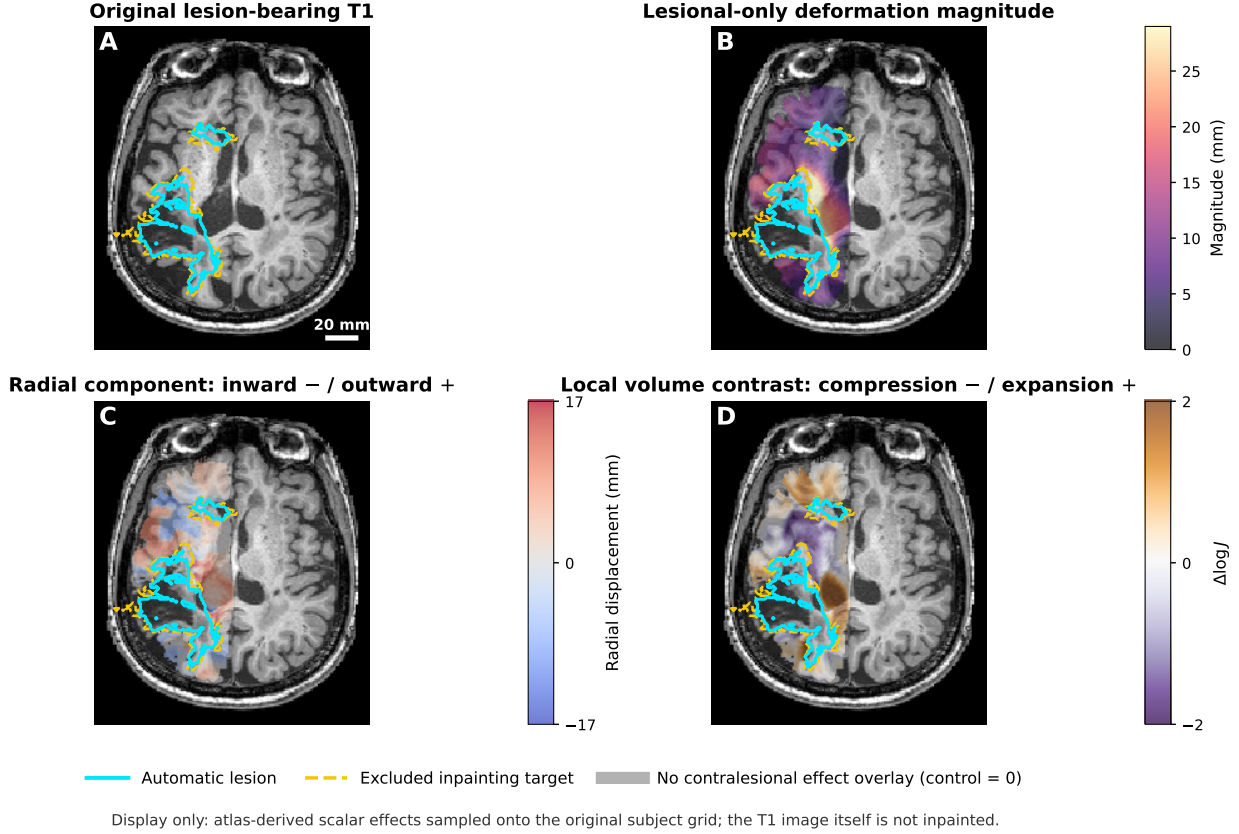


Figure 1: **Lesion-associated deformation construction in M2238.** (A) Acquisition-native T1 with lesion (cyan) and excluded inpainting target (yellow dashed). (B) Lesional-only deformation magnitude. (C) Radial displacement (negative inward, positive outward). (D) Lesional-minus-mirrored contralesional log-Jacobian. This preselected development case illustrates field construction and is not subject-specific evidence of pressure.

(IQR 2.95–4.48). Median direct-versus-inpainted registration sensitivity was 4.79 mm (IQR 3.64–6.23), giving a proxy-to-sensitivity ratio of 1.34 (IQR 1.00–1.71). Thus, the deformation scale was comparable to its preprocessing sensitivity, reinforcing the need for cautious cross-sectional interpretation.

Figure 1 shows M2238, selected as a high-asymmetry development case for technical visualization rather than as a representative participant. The cyan lesion and dashed yellow exclusion target are displayed on the original lesion-preserving T1; effect maps contain no target voxels.

3.2 Held-out AQ prediction

The clinical-only model performed poorly (MAE 24.23; R^2 -0.020). The conventional lesion model achieved MAE 16.42, RMSE 20.36, and R^2 0.454. Adding deformation reduced MAE to 16.04, whereas the registration-QC sensitivity model yielded 16.00 (Table 1; Fig. 2).

The primary mean MAE advantage was 0.38 points (95% bootstrap CI -0.50 to 1.17), and 60.5% of participants had lower repeated-CV error. The adjusted signed-rank result was $p_H = 0.023$, but the mean-effect CI crossed zero. Under the prespecified decision rule, deformation therefore

Table 1: Out-of-fold performance averaged across 20 repeated outer-CV splits on the identical 210-participant cohort.

Model	MAE	RMSE	R^2	Pearson r
Clinical	24.23	27.83	-0.020	-0.132
Conventional lesion	16.42	20.36	0.454	0.675
Lesion + deformation	16.04	20.33	0.456	0.677
Deformation + registration QC	16.00	20.40	0.452	0.674
Lesion + uncertainty	15.71	20.11	0.468	0.688
Uncertainty + deformation	15.77	19.99	0.473	0.689

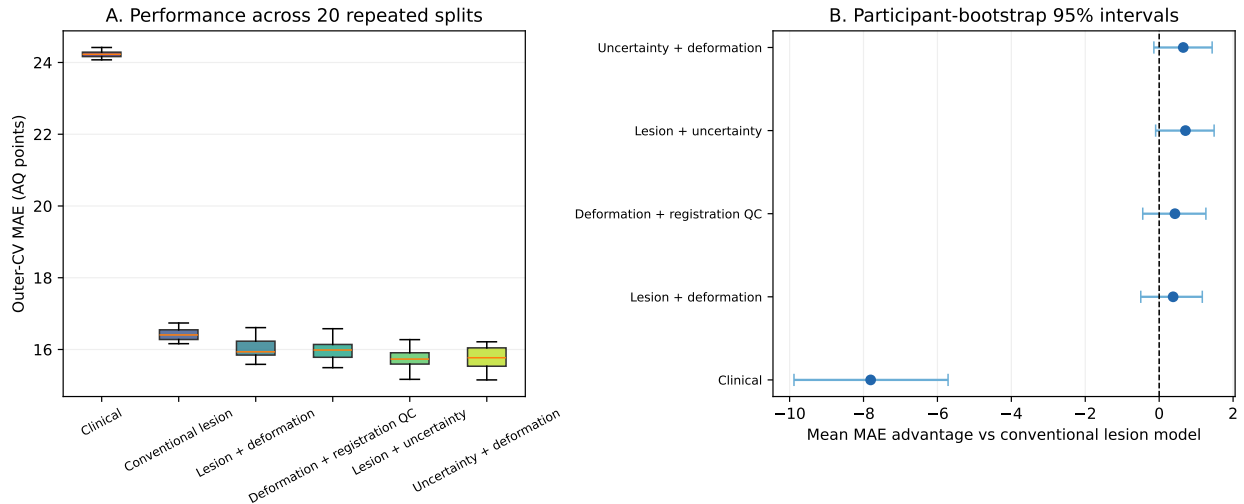


Figure 2: **Repeated nested-CV model comparison.** (A) Aggregate MAE for each complete outer-CV repeat. (B) Participant-level mean MAE advantage relative to the conventional lesion model; positive values favor the augmented model. Error bars are participant-bootstrap 95% CIs. Every augmented-model interval crosses zero; the deliberately under-specified clinical-only negative-control interval does not.

did not establish incremental predictive benefit. Adding registration-QC features gave a similar advantage of 0.43 points (CI -0.44 to 1.26).

The lesion-uncertainty model had MAE 15.71. Adding deformation to that model produced MAE 15.77 and a mean advantage of -0.06 points (CI -0.77 to 0.61 ; $p = 0.723$), providing no evidence of additive value.

4 Discussion

We constructed a fully auditable lesion-associated deformation proxy from registration fields in a large chronic stroke cohort. The method removes a contralaterally fitted affine component, subtracts mirrored control geometry, excludes generated targets and their homologs, and stores the effect only on the lesional side. These choices prevent a reflected mathematical copy from being mistaken for contralesional biology and provide explicit spatial support for every scalar summary.

The prediction experiment yielded a small apparent improvement over a conventional lesion model, but the primary participant-bootstrap interval included zero. The significant signed-rank result does

not contradict that conclusion: it concerns ranks of paired errors, whereas the prespecified effect was their mean difference. A skewed error distribution can produce these two outcomes. Reporting both, while conditioning the superiority claim on the mean-effect interval, avoids converting a distributional shift into an unsupported average-benefit claim.

Deformation added no value after soft lesion and boundary-ambiguity features. One explanation is informational overlap: deformation integrals and lesion uncertainty both scale with lesion extent and boundary complexity even after total volume and laterality are included. Another is measurement noise. Median proxy magnitude was only modestly larger than median direct-versus-inpainted registration sensitivity, so improved correspondences and residual algorithmic variation may obscure smaller behavioral signal.

The field is not physical ground-truth mass effect. Positive radial displacement does not establish ongoing pressure, and negative displacement can reflect collapse or ventricular expansion rather than reversal of acute displacement. The contralateral hemisphere is an internal reference, not a normative atlas; natural asymmetry and bilateral pathology violate its assumptions. The method is therefore restricted to strongly unilateral lesions and should be described as cross-sectional registration geometry.

Limitations include the absence of premorbid or acute reference scans, external validation, and expert ground truth for the deformation itself. The cohort is almost entirely left-lesioned because ARC focuses on aphasia. Upstream lesion delineation and inpainting can influence correspondences despite exclusion of target voxels. Repeated cross-validation reduces split dependence but does not replace validation in an independent cohort. Future work should include longitudinal deformation, non-inpainted and normative controls, tissue-specific summaries, uncertainty calibration, and preregistered external testing.

5 Conclusion

Contralateral normalization yields a transparent, lesional-only description of chronic stroke registration asymmetry. In ARC, deformation features produced a small reduction in held-out AQ error, but the primary confidence interval crossed zero and deformation added nothing after lesion uncertainty. The result supports the proxy as a research descriptor and does not support a claim of robust incremental prediction, pressure estimation, or causal mass effect.

Data and Code Availability

ARC is available through OpenNeuro as dataset ds004884, release 1.0.2 (<https://doi.org/10.18112/openneuro.ds004884.v1.0.2>). The complete analysis code, tests, frozen aggregate outputs, and manuscript sources accompany this submission. Participant-level derived images and joined clinical tables are not redistributed by this repository.

Declarations

Author identities, contributions, funding, acknowledgments, and competing interest statements are withheld in this blinded manuscript and must be supplied from the journal submission system.

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